Peer review of the Iowa data center nitrate study

Independent methodological and computational assessment

November 11 2026

Recommendation

Reject in its current empirical form. The principal outcome and weather variables are generated by the supplied code rather than obtained from observed water records. This is a foundational data-provenance failure that prevents interpretation of the reported causal effects, equivalence tests, and mechanism results. A corrected empirical paper requires rebuilding the dataset and re-estimating the analysis; editing the prose or replacing one estimator is insufficient.

Scope and successful checks

I reviewed the 17-page manuscript, accompanying media article, report, prompt and framing materials, 70-row facility inventory, and both Python scripts at commit 7e1f0e6d2a3c053a47ed9d90b45e894a9e02066a. I re-executed the code in an isolated output directory and independently retrieved EPA nitrate records. This assessment concerns the verifiable behavior of those materials and does not infer the intent of their author or generator.

The research question is relevant, and the proposed distinction between construction, operation, source water, and finished water could support a useful study. All 48 PWS identifiers match Iowa's public index. The facility CSV reproduces from the code, and the headline TWFE estimates reproduce to reported precision. Of 50 unique URLs embedded in the manuscript, 47 returned HTTP 200 and three returned HTTP 403 in this session. Link accessibility does not establish that an underlying source supports every cited numerical claim.

Major concern 1 Generated outcomes are presented as measurements

In 01_fetch_and_clean_data.py, build_pws_nitrate_panels starts a fixed random generator near line 609, assigns assumed weather values near lines 611–635, and computes nitrate from manually supplied system baselines, seasonal amplitudes, source sensitivities, and random noise near lines 647–714. Selected system-year overrides appear near lines 716–725. The code contains no parser that turns returned Water Quality Portal observations into the reported panel.

The external-file path only checks existence and logs "ingesting directly." A challenge file containing an extreme nitrate value did not alter any quarterly outcome: the panel hash was identical. The manuscript nevertheless describes a balanced observed monitoring panel on pages 1 and 6–7, and the media article presents the resulting estimates as an actual statewide finding.

Required remedy Remove all empirical effect and equivalence claims based on these values. Supply the raw downloaded records, metadata, extraction logs, sample-level cleaning code, and traceable joins from samples to analysis rows. The 912-row generated sample preview cannot serve as the underlying measurement archive.

Major concern 2 Uncertainty and estimator labels are invalid

In 02_did_equivalence_analysis.py, run_stacked_and_cs_did sets se_cs = se_stk * 1.04 and se_overall_cs = se_overall_stk * 1.03. These are not variance estimates for the cohort-time ATT. The same routine substitutes a stacked coefficient when a cohort-time estimate is missing. Pointwise intervals and equivalence tests based on these quantities have no justified sampling interpretation.

The cohort-time calculation residualizes outcomes using a pooled regression among never-treated observations. It is not shown to implement the conditional-parallel-trends estimator or its influence function described by Callaway and Sant'Anna. The stacked regression is labeled interchangeably as Sun–Abraham and stacked DiD, although the interaction-weighted estimator is not implemented. The routine also excludes the 2006 and 2007 treated cohorts without substantive justification.

Required remedy Describe the estimator actually used. Estimate cohort-time effects and their covariance using a validated implementation or a transparent paired-change estimator with a supply-cluster bootstrap that recomputes all stages. Report support and weights for each event time. Do not borrow uncertainty from a different estimator or replace missing event times with another method's result.

Major concern 3 Exposure definitions do not establish the pathways

The "same municipality" rule matches city strings rather than verified service boundaries. Distance buffers use hard-coded PWS coordinates with no demonstrated intake location accuracy. The hydrologic rule uses a manually supplied HUC10 lookup list. A shared watershed code is not proof of upstream connectivity, and surface watersheds do not establish groundwater transport to a deep well.

Purchased-water municipalities can share source concentrations and treatment decisions even when their local watersheds differ. That network can contaminate the treatment/control contrast and create dependence across county or HUC clusters. The manuscript does not establish which data centers buy water from which suppliers or where cooling return flows enter a receiving water.

Required remedy Provide a dated facility-to-source and facility-to-supplier crosswalk with documentary evidence. Separate a source-contamination estimand from a demand-mediated finished-water estimand. Use intake catchments and flow direction for surface water; aquifer and capture zones for groundwater; and observed purchase/withdrawal arrangements for demand. Treat proximity as exploratory if stronger links cannot be established.

Major concern 4 Timing and cumulative exposure are misclassified

Quarterly treatment indicators are based on calendar year even when quarter dates exist. This assigns treatment before a late-year construction or activation event. Mechanism variables assign all matched site acreage or power at the earliest event, incorporating future expansions into earlier exposures. Construction and operation are modeled separately without establishing how ongoing construction effects are distinguished from activation effects.

Required remedy Use verified lifecycle dates at their actual precision; carry interval uncertainty where necessary. Define campuses and expansions consistently. Distinguish approval, disturbance, first withdrawal, and later intensity changes. Reframe operation estimates as post-activation associations unless the design identifies an incremental operation effect beyond construction.

Major concern 5 The written results are not fully reproducible

Quantity	Written material	Reexecution
Construction TWFE	+0.0013	+0.00129
Operation TWFE	−0.0124	−0.01238
Group time overall ATT	−0.0212	−0.01627
Stacked static ATT in report	+0.0018	−0.00747
Treated systems within 10 km	21	31
Treated same municipality	15	19

Effect units are mg/L as N. The first two values reproduce to the displayed precision; the others require reconciliation. All effect results here are generated-data forensic outputs. The complete comparison tables accompany this review.

The facility CSV also sums to 6,606.5 MW across 20 county names, whereas the narrative describes 6,200.5 deduplicated MW and 18 counties. A deduplicated total could be defensible, but its campus/phase rules and transformation are not implemented or documented in the supplied output. Permitting effects are asserted in the abstract even though the supplied model specification list does not estimate a permitting treatment.

Required remedy Generate manuscript tables directly from analysis outputs, document every sample restriction, and reconcile project-level counts and capacities. Do not claim statewide completeness from a tracker headline count alone. Independently verify individual facility records and exact dates; generic tracker URLs do not validate all row-level fields.

Major concern 6 Equivalence and pretrend claims are overstated

Even on observed data, a TOST result would support equivalence only for a specified estimand, margin, and uncertainty model. It would not prove the null hypothesis beta = 0. The 0.25 mg/L margin is not justified beyond being a fraction of the regulatory limit. The same numeric margin is passed to the binary exceedance model, where it means 25 percentage points. The 0.05 SD margin depends on the generated outcome distribution.

The code sets pre_trends_parallel_confirmed to true when a pretrend p value exceeds 0.10. A low-powered failure to reject cannot confirm parallel trends. The reported post hoc MDE is a standard-error-based approximation and does not demonstrate the power of a realistic irregular monitoring design with shared suppliers and few treated clusters.

Required remedy Specify outcome-specific margins independently of the result; report concentration and probability effects in their correct units. Show event-time support, simultaneous uncertainty, and sensitivity to plausible trend deviations. Plan power using observed variance, missingness, temporal dependence, and the number of independent exposed source/supply clusters. Present nonsignificant pretrends as inconclusive diagnostics.

Major concern 7 The mechanism and public claims exceed the evidence

The precipitation and temperature coefficients are generated by construction of the outcome, not estimates using measured weather. The manuscript cannot use them to conclude that agricultural flushing rather than data centers caused the 2025–2026 crises. The media article's "flat zero," assertion that construction removes nitrogen, and attribution of more than 95% of nitrate to upstream agriculture require evidence separate from this pipeline.

Cooling's lack of direct nitrogen production does not exclude effects through construction, altered pumping or blending, withdrawal-induced flow changes, or downstream return flows. Conversely, water demand and nitrate concentration are distinct outcomes. A nitrate-equivalence result would not establish equivalence of treatment costs, capacity, or availability.

Required remedy Remove causal mechanism and policy-exoneration language unless supported by independent analyses. Use a neutral policy brief that distinguishes known facts, the unresolved question, and required evidence. Avoid treating nitrate below the MCL as proof of no exposure concern or a short-term null as evidence about cancer outcomes.

What the independent data retrieval establishes

EPA's public Six-Year Review 4 archive was downloadable in this session. It contains 28,103 Iowa-filtered nitrate records for 2012–2019. Restricting to usable routine community samples leaves 17,011 records from 845 systems. Source-stage metadata are unresolved for 96.9% of those records; only 521 are classified as finished or entry and one as raw using the available codes. These data cannot simply replace the purported balanced 2005–2026 panel. Among the original 48 systems, 38 have records in this release; absence of a record does not demonstrate absence of monitoring.

I retained unknown stages, sample-specific censoring intervals, and missing quarters. The replacement pipeline produces descriptive coverage results and refuses causal estimation without a reviewed exposure crosswalk. This is the appropriate stopping point for the currently verified inputs, while targeted data acquisition continues through the supplied code and request specification.

Presentation and reproducibility revisions

- Replace the unsupported institutional byline and all author or preprint placeholders with verified author information before submission or dissemination.
- Separate raw, entry, and distribution-water observations; report the sampling unit, weighting convention, censoring burden, and points per system-period.
- Explain whether maxima are sample maxima or measured continuous peaks, and assess sampling-frequency sensitivity.
- Distinguish an observed HTTP or form error from an assumed CAPTCHA or an inability to obtain any data. Retain failed attempts in an acquisition log.
- Keep generated demonstration outputs visibly segregated from observed records. Release hashes, executable code, data dictionaries, and machine-generated result tables.

Overall assessment

The question merits a new study. The submitted empirical conclusion is not supported by the supplied project. Rebuilding from observed data and verified source connections is necessary before either a harmful-effect or negligible-effect conclusion can be evaluated.